

ADSR-Controller

An external GATE input for the ADSR envelope generators is frequently requested, particularly when the FORMANT system is intended for use in conjunction with sequencers. Via a "Repeat" input, the ADSR function can be triggered repeatedly in sync with an LFO square wave signal. The unit's integrated delay section allows for the time-delayed routing of an incoming Gate pulse to the "triggered" elements. ADSR envelope generators serve primarily to control the timbral dynamics of VCFs and the amplitude contour of VCAs. Within the context of the FORMANT series, it was originally proposed that the ADSRs be hardwired internally; consequently, they do not possess their own dedicated inputs. If one wishes to operate the FORMANT in conjunction with a sequencer—for instance—one encounters difficulties in routing the Gate signals generated by the sequencer to the envelope generators. In addition, the ADSR Controller offers the capability to derive the Gate signal from the square wave output of an LFO (which may, of course, be voltage-controlled—i.e., a VC-LFO or LF-VCO), or alternatively, to periodically interrupt a Gate signal received from either an internal or external source, thereby causing the ADSR envelope to repeat in time with the square wave signal. As a unique feature, the ADSR Controller provides the option to delay the Gate signal for a specific envelope generator—a function that can be enabled/disabled (i.e., pre-programmed)—within a time range of 10 ms to approximately 5 seconds. This enables the staggered triggering of individual envelope generators, thereby allowing for the creation of "chorus-like" effects using a single keyboard (provided that multiple independent signal paths are available).

The Circuit

The circuit principle is illustrated in Figure 1. Figure 2 shows the complete schematic of the "Control Unit."

When switch S1 is in the "KV-GATE" position, and S2 and S3 are in the "NORMAL" and "OFF" positions respectively, the gate signal from the interface receiver passes unaltered to the corresponding envelope generator. Switch S1 allows for switching to an external gate signal (+5V = ON, 0V = OFF).

The square-wave signal from an LFO—used for the repeat effect of the envelope function—undergoes a level shift of +2.3 V via OpAmp IC2. When switch S2 is in the "AUTO-REPEAT" position, the connected ADSR module tracks the clock frequency of the LFO. In the "KB-REPEAT" position, the output signal from comparator IC5 is routed via R9 and P1 to the (internally wired) ADSR input. The inverted Gate signal—shifted in level by 5 V—serves as the reference voltage at the comparator's inverting input. The unmodified square-wave signal (upper plateau)—which is simultaneously present at the module's input—is fed to the non-inverting input. Thus, the Gate signal—repeating in sync with the LFO signal—follows either the key press or the time constant of a sequencer's trigger output.

The delay circuit within the ADSR controller is very similar in design to the actual FORMANT ADSR circuit. The function of the delay unit is briefly explained below: The gate signal present at the input of the delay circuit uses its positive edge to trigger the monostable multivibrator (monoflop) formed by transistors T1 and T2; this, in turn, delivers a set pulse to the flip-flop formed by transistors T4 and T5. The flip-flop is set, causing T5 to conduct and thereby cut off T4. Since T4 is now cut off, its collector potential rises to +15V; T3 begins to conduct, thereby enabling the "delay charging path." This path is fed by a +6V voltage source (IC1) derived from the gate signal. Consequently, the tantalum capacitor C2 charges up over a delay period that is adjustable via potentiometer P2. The voltage across C2 is buffered at low impedance by an operational amplifier (IC4). The relatively high input impedance of IC4 ensures that practically the entire output current from IC1 flows into capacitor C2. From the output of IC4, the signal proceeds to the non-inverting input of the comparator IC3. As soon as the voltage across capacitor C2 reaches the comparator's switching threshold (approx. +5V), the output voltage of IC3 switches abruptly from approx. -12V to -14V up to +12V to +14V; this action resets the T4/T5 flip-flop via diode D3 and resistor R30. Consequently, T3 cuts off, thereby terminating the charging phase of C2 (the "Delay Time"). The voltage across C2 is now held—or stored—until the "end" of the gate signal at the input of the delay circuit. Upon the falling edge of the gate signal, the output voltage of IC1 also drops to 0V, allowing the voltage stored in C2 to discharge through diode D1 and resistor R29. With R29, the discharge constant is set to approximately 3 ms. The delay circuit is now ready to delay the next Gate signal. Using P3, the positive output voltage at comparator IC3 is adjusted to approximately +5 V. Any negative output voltage is clamped to ground potential via D5. Consequently, a "proper" Gate signal is now available at the output of the delay circuit. Switch S3 allows for the pre-programming of a specific delay time, enabling the Gate signal to be delayed at precisely the right moment and for the desired duration.

Component Selection

As a general rule—as has been mentioned several times throughout the FORMANT series—one should use only components from reputable manufacturers. For C2, it is absolutely essential to use a tantalum electrolytic capacitor. For the TUN transistors T1 through T5, C-type transistors should be used (e.g., BC 550C).

Construction and Calibration

The PCB layout and component placement diagram are shown in Figure 3. Construction should present no difficulties. As a general rule, a dedicated ADSR controller module should be provided for each ADSR envelope generator. Wiring is carried out internally. Alternatively, an output jack could be provided on the ADSR controller itself. In this case, the ADSR module would need to be fitted with a switched input jack, through which the Gate line is routed. When the ADSR controller output is connected to the ADSR input, the keyboard Gate signal from the interface receiver is automatically disconnected. In this way, the number of controller modules can be reduced. A useful rule of thumb is to provide half the number of ADSR modules.

The only calibration points are trimmers P1 and P3, which are used to set the "amplitude" of the Gate signal to approximately +4.7 V to +5.0 V. The easiest way to adjust P1 is to set the LFO to its maximum period duration (i.e., minimum frequency) while simultaneously holding down a key (with switch S1 in the KB-GATE position). P1 can then be easily adjusted to the value required to reliably "trigger" the ADSR envelope generators. P3 is likewise set to +4.7 V to +5.0 V while a key is held down and the Delay time is set to its minimum.

Bill of Materials for Figure 2

Resistors (Carbon Film, 5%):

R1, R2, R4, R8, R10, R13, R14, R21, R26, R27 = 100 k Ω

R3, R7, R18 = 4.7 k Ω

R5, R17, R23 = 10 k Ω

R6, R24 = 27 k Ω

R9, R19 = 1 k Ω

R11, R16, R22, R30 = 33 k Ω

R12, R15 = 5.6 k Ω

R20 = 220 Ω

R25, R28 = 6.8 k Ω

R29 = 330 Ω

Potentiometers:

P1, P3 = 1 k Ω (Trimmer)

P2 = 1 M Ω (Logarithmic)

Capacitors:

C1 = 10 nF (MKH, MKM)

C2 = 2.2 μ F / 16 V, Tantalum

C3, C4 = 10 μ F / 25 V

Semiconductors:

D1, D2, D3, D6 = DUS

D4, D5 = DUG

T1...T5 = TUN (e.g., BC 550C)

IC1...IC6 = μ A741C (Mini-DIP)

Miscellaneous:

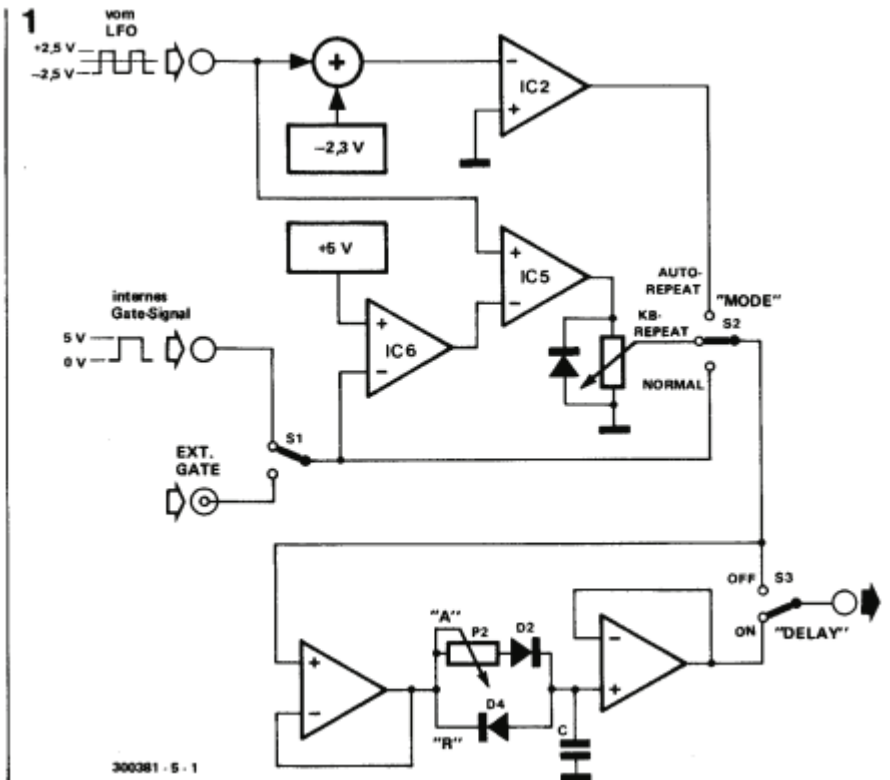
S1, S3 = Miniature Toggle Switch, SPDT (1x UM)

S2 = 3-pole Rotary Switch

3 x 3.5 mm Jack Sockets
2 x Control Knobs for 6 mm Shafts
1 x 31-pin Header Strip or Solder Pins

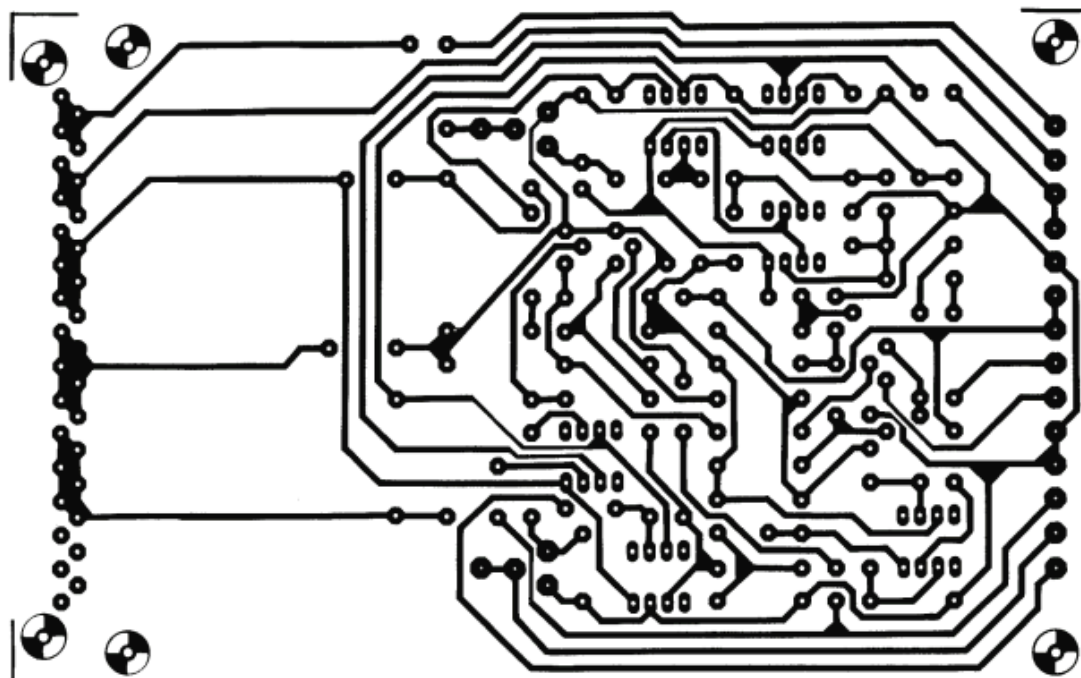
Figures

1. Simplified circuit principle of the ADSR controller. The delay line corresponds to the AR section of the FORMANT ADSR with $D = 0$.
2. Complete schematic of the FORMANT ADSR controller module.
3. PCB layout and component placement guide for the ADSR controller.
4. Suggested front panel design for the ADSR controller.
5. Basic circuit of the ADSR controller in the KB-REPEAT position.
6. Wiring diagram connecting the PCB to the front panel components

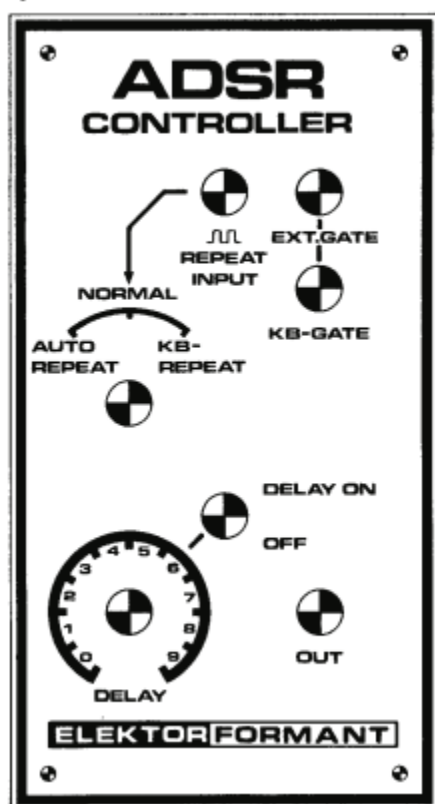




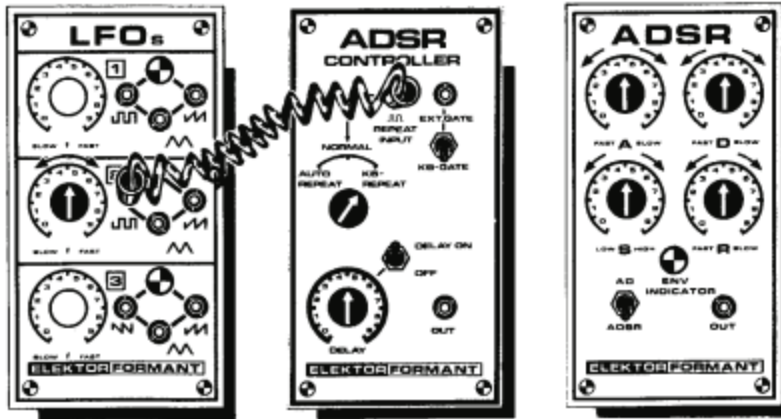
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